

Two partial balayage principles for weak-type estimates

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Abstract

We formulate two principles for weak-type (1,1) estimates. The first applies to linear operators through a locality identity; the second applies to maximal operators through comparison with a positive kernel. We explain the principles and apply them to obtain the following bounds.

Operator	Upper bound for the weak-type (1,1) constant obtained by the principles
Riesz transform	2
Beurling–Ahlfors transform	2
Second-order Riesz transform, $n = 2$	$\frac{3\sqrt{6}}{4}$
Second-order Riesz transform, $n \geq 2$	$\frac{1}{a_n} + \frac{(n-1)a_n}{n-a_n^2}, a_n := \sqrt{\frac{2n}{n+1+\sqrt{(n+1)^2+4(n-2)}}$
Traceless second-order Riesz transform	$2\sqrt{1-n^{-1}}$
Leray and gradient projections	≈ 1.805
Centered Hardy–Littlewood maximal operator, $n = 1$	2
Centered Hardy–Littlewood maximal operator for balls, $n = 2$	e
Centered Hardy–Littlewood maximal operator for axis-parallel squares, $n = 2$	< 3.616
Centered Hardy–Littlewood maximal operator for balls, $n \geq 3$	$(n/2)^{n/(n-2)}$
Poisson maximal operator, $n = 1$	$1 + \frac{2}{\pi} \left(\frac{\sqrt{5}}{3} - \arctan \frac{2}{\sqrt{5}} \right) \approx 1.010$
Poisson maximal operator, $n = 2$	≈ 1.021
Poisson maximal operator	$\frac{\Gamma((n+1)/2)}{\sqrt{\pi} \Gamma(n/2)} \left[\frac{2b^{n/2}}{n(1+a)^{(n+1)/2}} + \int_b^\infty \frac{z^{n/2-1}}{(1+z)^{(n+1)/2}} dz \right]$
Heat maximal operator, $n = 1$	≈ 1.037087
Heat maximal operator, $n = 2$	≈ 1.094052
Heat maximal operator	$\frac{1}{\Gamma(n/2)} \left[\frac{2(\rho_n a)^{n/2} e^{-a}}{n} + \int_{\rho_n a}^\infty z^{n/2-1} e^{-z} dz \right],$ $\rho_n = \begin{cases} (n/2)^{2/(n-2)}, & n \neq 2, \\ e, & n = 2. \end{cases}$ Here a is the unique nonzero solution in $(n/(2\rho_n), n/2)$ of $e^{-(\rho_n-1)a} = 1 - \frac{2a}{n}$.

1 Linear principle

1.1 The decomposition

The following theorem extends [2, Theorem 1.2] to finite-dimensional Hilbert-valued functions and includes the endpoint $\alpha = 2$.

Theorem 1.1. *Let E be a finite-dimensional real or complex Hilbert space and $0 < \alpha \leq 2$. For $f \in L^1(\mathbb{R}^n; E) \cap L^2(\mathbb{R}^n; E)$ and $\kappa > 0$, there exist $u \in H^\alpha(\mathbb{R}^n; E) \cap L^1(\mathbb{R}^n; E)$ and $\mu \in L^1(\mathbb{R}^n; E) \cap L^\infty(\mathbb{R}^n; E)$ with*

- $f = \mu + (-\Delta)^{\alpha/2}u$;
- $\|\mu\|_\infty \leq \kappa$ and $\|\mu\|_1 \leq \|f\|_1$;
- $\mu = \kappa u/|u|_E$ almost everywhere on $\Omega := \{|u|_E > 0\}$;
- $\kappa|\Omega| \leq \|f\|_1$.

For $\alpha = 2$, both ∇u and D^2u vanish almost everywhere on Ω^c , and consequently $f = \mu$ there.

Proof. Replace the scalar absolute value in the variational argument of [2, Section 4] by the Hilbert norm, and minimize

$$J(v) = \frac{1}{2} \|(-\Delta)^{\alpha/4}v\|_2^2 + \kappa \|v\|_1 - \operatorname{Re} \int \langle f, v \rangle_E$$

over $H^{\alpha/2} \cap L^1$. The same compactness argument gives a unique minimizer u . At $\alpha = 2$, the energy is $\frac{1}{2} \|\nabla v\|_2^2$.

The *Euler condition* means that no perturbation $u + th$ lowers J to first order; t is real even when E is complex. At $u \neq 0$, the norm has directional derivative $\operatorname{Re} \langle u/|u|_E, h \rangle_E$; at zero its one-sided directional derivative is $|h|_E$. Minimality in both directions gives

$$(-\Delta)^{\alpha/2}u + \mu = f, \quad |\mu|_E \leq \kappa, \quad \mu = \kappa u/|u|_E \quad \text{on } \Omega.$$

In the nonnegative scalar case the saturated value is κ . Here it is a vector of length κ pointing along u .

For the L^1 estimate, choose $w = u/|u|_E$ on Ω and $w = \mu/|\mu|_E$ elsewhere, taking zero when both vanish. The positive averaging semigroup $P_t = e^{-t(-\Delta)^{\alpha/2}}$ satisfies

$$|u|_E - P_t|u|_E \leq \operatorname{Re} \langle w, u - P_t u \rangle_E.$$

Divide by t and pass to the limit against nonnegative smooth test functions. Since $(u - P_t u)/t \rightarrow f - \mu$ in L^2_{loc} , this gives

$$(-\Delta)^{\alpha/2}|u|_E \leq |f|_E - |\mu|_E \quad \text{in distributions.}$$

Test with $\chi_R = \chi(\cdot/R)$, where $0 \leq \chi \leq 1$ is smooth, compactly supported, and equals one near zero. Since $\|(-\Delta)^{\alpha/2}\chi_R\|_\infty = O(R^{-\alpha})$, letting $R \rightarrow \infty$ gives $\|\mu\|_1 \leq \|f\|_1$. Saturation then gives $\kappa|\Omega| \leq \|f\|_1$.

Now $\mu \in L^1 \cap L^\infty \subset L^2$, so Plancherel gives $u \in H^\alpha$. At $\alpha = 2$, applying the Sobolev zero-set property to u and then to its first derivatives gives $\nabla u = D^2u = 0$ on Ω^c , hence $\mu = f$ there. For scalar $f \geq 0$, replacing u by $|u|$ does not increase J , so $u \geq 0$. Testing the decomposition with χ_R also gives $\int \mu = \int f$, which together with the L^1 bound implies $\mu \geq 0$. $\square$

For nonnegative scalar data, this is the partial balayage construction: f is redistributed into the capped density μ , while the odometer u records the redistribution through $(-\Delta)^{\alpha/2}u = f - \mu$. We now apply Theorem 1.1 to obtain our first principle.

Principle 1 (Linear). *Let E and F be finite-dimensional real or complex Hilbert spaces, and let $T : L^2(\mathbb{R}^n; E) \rightarrow L^2(\mathbb{R}^n; F)$ be a bounded linear operator. Suppose that $\|T\|_{2 \rightarrow 2} \leq M$ and, for some $0 < \alpha \leq 2$, the composition $T(-\Delta)^{\alpha/2}$ is local in the following sense: for every $u \in H^\alpha(\mathbb{R}^n; E)$, $T(-\Delta)^{\alpha/2}u = 0$ almost everywhere on $\{u = 0\}$. Then $\|T\|_{1 \rightarrow 1, \infty} \leq 2M$.*

In the applications below, $T(-\Delta)^{\alpha/2}$ is a differential operator; the required vanishing follows from the Sobolev zero-set property.

Proof. For $f \in L^1 \cap L^2$, use the decomposition $f = \mu + (-\Delta)^{\alpha/2}u$. Outside Ω one has $Tf = T\mu + T(-\Delta)^{\alpha/2}u = T\mu$ by locality, so Chebyshev's inequality gives

$$\lambda |\{ |Tf| > \lambda \}| \leq \lambda |\Omega| + \lambda^{-1} \|T\mu\|_2^2 \leq \left(\frac{\lambda}{\kappa} + \frac{M^2 \kappa}{\lambda} \right) \|f\|_1.$$

Here $\|\mu\|_2^2 \leq \kappa \|\mu\|_1 \leq \kappa \|f\|_1$. For $M > 0$, choose $\kappa = \lambda/M$. The case $M = 0$ is trivial. The estimate extends to L^1 by density. $\square$

An identity component can improve the bound.

Principle 1.2 (Variant of Principle 1). *Suppose $E = F$ and $T = cI + S$ for some $c \neq 0$, with $\|S\|_{2 \rightarrow 2} \leq M$. Assume the same locality condition as in Principle 1. Then*

$$\|T\|_{1 \rightarrow 1, \infty} \leq \inf_{0 < a < 1/|c|} \left\{ \frac{1}{a} + \frac{M^2 a}{(1 - |c|a)^2} \right\}.$$

Proof. Choose $0 < \kappa < \lambda/|c|$ and write $f = \mu + (-\Delta)^{\alpha/2}u$ using the decomposition. Outside Ω one has

$$|Tf| = |T\mu| = |c\mu + S\mu| \leq |c|\kappa + |S\mu|.$$

Hence, up to a null set,

$$\{|Tf| > \lambda\} \subseteq \Omega \cup \{|S\mu| > \lambda - |c|\kappa\}.$$

The restriction on κ makes the latter threshold positive. Applying Chebyshev's inequality gives

$$\lambda |\{|Tf| > \lambda\}| \leq \lambda |\Omega| + \frac{\lambda \|S\mu\|_2^2}{(\lambda - |c|\kappa)^2} \leq \left(\frac{\lambda}{\kappa} + \frac{\lambda M^2 \kappa}{(\lambda - |c|\kappa)^2} \right) \|f\|_1.$$

Here $\|S\mu\|_2^2 \leq M^2 \|\mu\|_2^2 \leq M^2 \kappa \|\mu\|_1 \leq M^2 \kappa \|f\|_1$. Set $\kappa = a\lambda$, where $0 < a < 1/|c|$. The estimate becomes

$$\lambda |\{|Tf| > \lambda\}| \leq \left(\frac{1}{a} + \frac{M^2 a}{(1 - |c|a)^2} \right) \|f\|_1.$$

$\square$

2 Applications of the linear principle

2.1 Riesz transform

Denote the Riesz transform by $R = \nabla(-\Delta)^{-1/2}$. Its Fourier multiplier is $i\xi/|\xi|$, so Plancherel gives $\|R\|_{2 \rightarrow 2} = 1$. $R(-\Delta)^{1/2}u = \nabla u$. Principle 1 therefore gives $\|R\|_{1 \rightarrow 1, \infty} \leq 2$. This is exactly Theorem 1.1 in [2].

2.2 Beurling–Ahlfors transform

The Beurling–Ahlfors transform is

$$Bf(z) = -\frac{1}{\pi} \text{p. v.} \int_{\mathbb{C}} \frac{f(w)}{(z-w)^2} dA(w), \quad \widehat{Bf}(\xi) = \frac{(\xi_1 - i\xi_2)^2}{|\xi|^2} \widehat{f}(\xi).$$

Its multiplier has modulus one and thus $\|B\|_{2 \rightarrow 2} = 1$. With $D = \partial_1 - i\partial_2$, $B(-\Delta u) = -D^2 u$. Principle 1 gives $\|B\|_{1 \rightarrow 1, \infty} \leq 2$.

Classical Calderón–Zygmund theory gives weak-type $(1, 1)$ boundedness; Astala–Iwaniec–Martin record the explicit upper bound 30 [3, Theorem 4.5.2]. Guerra [4, Corollary 1.2] proved the sharp constant 2 for real-valued inputs using quasiconvexity and Morse theory.

2.3 Second-order Riesz transforms

For scalar f , define the matrix-valued second-order Riesz transform $\mathcal{H}_n f = (R_j R_k f)_{j,k} = D^2(-\Delta)^{-1} f$. The multiplier of $\mathcal{H}_n$ is $-\xi \otimes \xi / |\xi|^2$. With the Frobenius norm, $\|\mathcal{H}_n\|_{2 \rightarrow 2} = 1$. Define its traceless part by $\mathcal{H}_{0,n} f = \mathcal{H}_n f + \frac{f}{n} I$. Orthogonality to the identity matrix gives

$$|\mathcal{H}_n f|_F^2 = |\mathcal{H}_{0,n} f|_F^2 + \frac{|f|^2}{n}, \quad \|\mathcal{H}_{0,n} f\|_2^2 = \left(1 - \frac{1}{n}\right) \|f\|_2^2.$$

Both compositions are local:

$$\mathcal{H}_n(-\Delta u) = D^2 u, \quad \mathcal{H}_{0,n}(-\Delta u) = D^2 u - \frac{\Delta u}{n} I.$$

Principle 1 therefore gives $\|\mathcal{H}_n\|_{1 \rightarrow 1, \infty} \leq 2$ and $\|\mathcal{H}_{0,n}\|_{1 \rightarrow 1, \infty} \leq 2\sqrt{1 - 1/n}$. The trace identity improves the first bound.

Corollary 2.1. *For $n \geq 2$,*

$$\|\mathcal{H}_n\|_{1 \rightarrow 1, \infty} \leq \min_{0 < a < \sqrt{n}} \left(\frac{1}{a} + \frac{(n-1)a}{n-a^2} \right) = \frac{1}{a_n} + \frac{(n-1)a_n}{n-a_n^2}, \quad (1)$$

where

$$a_n^2 = \frac{2n}{n+1 + \sqrt{(n+1)^2 + 4(n-2)}}.$$

In dimension two this upper bound is $3\sqrt{6}/4 = 1.837117307087\dots$. For every $n \geq 2$ it is less than 2, and as $n \rightarrow \infty$ it equals $2 - n^{-2} + 2n^{-3} + O(n^{-4})$.

Proof. The proof is similar to that of Principle 1.2. Write $f = \mu - \Delta u$ with $\text{cap } \kappa$. Outside $\Omega = \{|u| > 0\}$, locality and trace orthogonality imply

$$|\mathcal{H}_n f|_F^2 = |\mathcal{H}_n \mu|_F^2 = |\mathcal{H}_{0,n} \mu|_F^2 + \frac{|\mu|^2}{n} \leq |\mathcal{H}_{0,n} \mu|_F^2 + \frac{\kappa^2}{n}.$$

For $0 < \kappa < \sqrt{n} \lambda$, Chebyshev’s inequality yields

$$\lambda |\{|\mathcal{H}_n f|_F > \lambda\}| \leq \lambda |\Omega| + \frac{\lambda \|\mathcal{H}_{0,n} \mu\|_2^2}{\lambda^2 - \kappa^2/n} \leq \left(\frac{\lambda}{\kappa} + \frac{\lambda(n-1)\kappa}{n\lambda^2 - \kappa^2} \right) \|f\|_1.$$

Set $a = \kappa/\lambda$. Differentiating the resulting expression gives

$$(n-2)a^4 + n(n+1)a^2 - n^2 = 0,$$

whose unique positive solution gives the stated minimum. $\square$

For real-valued f in dimension two,

$$Bf = -(\mathcal{H}_2 f)_{11} + (\mathcal{H}_2 f)_{22} + 2i(\mathcal{H}_2 f)_{12}, \quad |\mathcal{H}_2 f|_F^2 = \frac{|f|^2 + |Bf|^2}{2}, \quad |\mathcal{H}_{0,2} f|_F = \frac{|Bf|}{\sqrt{2}}.$$

Guerra's quasiconvexity theorem and sharp Beurling–Ahlfors estimate [4, Theorem 1.1 and Corollary 1.2] imply, for real-valued inputs,

$$\|\mathcal{H}_2\|_{1 \rightarrow 1, \infty} = \|\mathcal{H}_{0,2}\|_{1 \rightarrow 1, \infty} = \sqrt{2}.$$

We give the deduction in Appendix A. Thus the method we use gives a sharp bound to $\mathcal{H}_{0,2}$ but not to $\mathcal{H}_2$.

2.4 Projections

For $n \geq 2$, the Helmholtz–Leray projection and its complement have multipliers

$$\widehat{\mathbb{P}}f(\xi) = \left(I - \frac{\xi \otimes \xi}{|\xi|^2} \right) \widehat{f}(\xi), \quad \mathbb{Q} = I - \mathbb{P}.$$

The fields carry the pointwise Euclidean norm (Hermitian norm for complex fields). These operators extract their divergence-free and gradient parts. Classical singular-integral theory gives weak $(1, 1)$ boundedness in each dimension [5]. To obtain an explicit uniform bound, use the local identities:

$$\mathbb{P}(-\Delta u) = -\Delta u + \nabla \operatorname{div} u, \quad \mathbb{Q}(-\Delta u) = -\nabla \operatorname{div} u.$$

For either projection T , write $T = \frac{1}{2}I + S$. The Fourier multipliers project onto the direction of ξ and its orthogonal complement, so Plancherel's theorem gives $Tf \perp (I - T)f$ in L^2 . Hence

$$\|Sf\|_2^2 = \frac{1}{4}\|Tf - (I - T)f\|_2^2 = \frac{1}{4}(\|Tf\|_2^2 + \|(I - T)f\|_2^2) = \frac{1}{4}\|f\|_2^2.$$

Thus orthogonality gives $\|S\|_{2 \rightarrow 2} = 1/2$. Principle 1.2 gives the dimension-free bound

$$\|T\|_{1 \rightarrow 1, \infty} \leq \min_{0 < a < 2} \left(\frac{1}{a} + \frac{a}{(2 - a)^2} \right) = 1.805359306638 \dots$$

The minimizing a_* is the unique root in $(0, 1)$ of

$$a_*^3 - 2a_*^2 + 6a_* - 4 = 0, \quad \|T\|_{1 \rightarrow 1, \infty} \leq \frac{1}{a_*} + \frac{a_*}{(2 - a_*)^2}.$$

The gain comes from using pointwise control of the identity component and the smaller L^2 norm of the remainder.

3 Maximal principle

3.1 The decomposition

Let $\mathcal{A}$ be the generator of a conservative, symmetric Markov convolution semigroup on $\mathbb{R}^n$. This means

$$P_t g(x) = \int_{\mathbb{R}^n} g(x - y) d\nu_t(y),$$

where ν_t are probability measures invariant under $y \mapsto -y$, satisfying $\nu_s * \nu_t = \nu_{s+t}$ and $\nu_t \rightarrow \delta_0$ as $t \downarrow 0$. Thus P_t preserves positivity, constants, and mass, and is self-adjoint on L^2 . The generator is

$$\mathcal{A}g = \lim_{t \downarrow 0} \frac{P_t g - g}{t} \quad \text{in } L^2, \quad g \in \text{Dom}(\mathcal{A}),$$

with Fourier symbol $-\psi$, where $\psi \geq 0$. We assume that, for every bounded, compactly supported $f \geq 0$ and $\kappa > 0$, there exist $u \geq 0$ in $L^1 \cap \text{Dom}(\mathcal{A})$ and a density σ such that

$$f = \sigma - \mathcal{A}u, \quad 0 \leq \sigma \leq \kappa, \quad \sigma = \kappa \text{ on } D = \{u > 0\}, \quad \int \sigma = \int f.$$

Here $\text{Dom}(\mathcal{A})$ is the L^2 operator domain. In particular, $|D| \leq \|f\|_1/\kappa$. This is the partial balayage decomposition needed below. It holds for Δ by Theorem 1.1, and $\mathcal{A}_\alpha = -\sum_j |D_j|^\alpha$, $0 < \alpha < 2$, by the same variational argument: their energy is comparable to the fractional Sobolev energy, their semigroups preserve positivity and mass, and $\|\mathcal{A}_\alpha \chi(\cdot/R)\|_\infty = O(R^{-\alpha})$. The details are recorded in [8].

3.2 The estimate

Principle 2 (Maximal). *Suppose $\mathcal{A}$ is homogeneous of order $\alpha > 0$ and admits the decomposition above. Let $K \geq 0$ be an even L^1 kernel satisfying*

$$\mathcal{A}K \geq 0 \quad \text{in distributions on } \mathbb{R}^n \setminus \{0\}.$$

For $K_r(x) = r^{-n}K(x/r)$, define

$$M_K f(x) = \sup_{r>0} (K_r * |f|)(x).$$

Then

$$\|M_K\|_{1 \rightarrow 1, \infty} \leq \int_{\mathbb{R}^n} K.$$

Any measurable operator dominated in absolute value by M_K satisfies the same estimate.

Proof. We give an outline; details of the kernel argument for the coordinate-stable case can be found in [6, Lemma 3]. Write $M = \int K$ and assume $M > 0$, since $K = 0$ otherwise. For $\lambda > 0$ and bounded, compactly supported $f \geq 0$, take the decomposition at height $\kappa = \lambda/M$. It suffices to show that every $K_r * f$ is at most λ outside the same set $D = \{u > 0\}$.

Put $\mathcal{B}v = K * \mathcal{A}v$. The condition $\mathcal{A}K \geq 0$ away from the origin, together with a cutoff argument at the origin, gives the positive minimum principle for $\mathcal{B}$: if $\varphi \in C_c^\infty$ is nonnegative and $\varphi(x) = 0$, then $\mathcal{B}\varphi(x) \geq 0$. Courrège's theorem [7] then identifies $\mathcal{B}$ as a conservative, symmetric Markov generator. Its symbol is $-\psi\hat{K}$, with $0 \leq \psi\hat{K} \leq M\psi$, so u lies in its operator domain.

The Markov property gives $\mathcal{B}u \geq 0$ almost everywhere on D^c . Indeed, for $0 \leq \varphi \in C_c^\infty$, set $h_\varepsilon = (\varepsilon\varphi - u)_+$. The Dirichlet form satisfies

$$-\langle \mathcal{B}u, h_\varepsilon \rangle \leq -\varepsilon \langle \mathcal{B}\varphi, h_\varepsilon \rangle.$$

Divide by ε and let $\varepsilon \downarrow 0$, using $h_\varepsilon/\varepsilon \rightarrow \varphi \mathbf{1}_{D^c}$ and $\|h_\varepsilon\|_2 \leq \varepsilon\|\varphi\|_2$. Consequently,

$$K * f = K * \sigma - \mathcal{B}u \leq \kappa \int K = \lambda \quad \text{almost everywhere on } D^c.$$

Homogeneity gives $\mathcal{A}K_r = r^{-n-\alpha}(\mathcal{A}K)(\cdot/r) \geq 0$ away from zero, so the same argument applies to every dilation. Continuity of $r \mapsto K_r$ in L^1 extends the conclusion from rational scales to all scales outside one null set. Hence

$$\lambda|\{M_K f > \lambda\}| \leq \lambda|D| \leq M\|f\|_1.$$

Monotone approximation gives the result for general f . $\square$

Corollary 3.1. *For $\mathcal{A}$ as in Principle 2 and an even nonnegative $\phi \in L^1$, let*

$$M_\phi f = \sup_{r>0} \phi_r * |f|, \quad \phi_r(x) = r^{-n}\phi(x/r).$$

Then

$$\|M_\phi\|_{1 \rightarrow 1, \infty} \leq C_{\mathcal{A}}(\phi) := \inf \left\{ \int K : K \in L^1 \text{ is even, } K \geq \phi, \mathcal{A}K \geq 0 \text{ on } \mathbb{R}^n \setminus \{0\} \right\}.$$

Proof. Since $M_\phi \leq M_K$ for every admissible K , Principle 2 applies; take the infimum of the resulting bounds. $\square$

4 Applications of the maximal principle

4.1 Intervals

The normalized tent kernel is the simplest exact example of the maximal principle:

$$\tau(x) = (1 - |x|)_+, \quad \int_{\mathbb{R}} \tau = 1, \quad \tau'' = \delta_{-1} + \delta_1 - 2\delta_0.$$

Thus its dilation maximal operator

$$T_* f(x) = \sup_{r>0} (\tau_r * |f|)(x), \quad \tau_r(x) = r^{-1}\tau(x/r),$$

has weak-type constant at most one. For the centered Hardy–Littlewood maximal operator on intervals,

$$M_c f(x) = \sup_{r>0} \frac{1}{2r} \int_{x-r}^{x+r} |f(y)| dy,$$

the majorant

$$K(x) = (1 - |x|/2)_+ \geq \frac{1}{2} \mathbf{1}_{[-1,1]}(x), \quad \int K = 2,$$

gives $\|M_c\|_{1 \rightarrow 1, \infty} \leq 2$. This majorant attains $C_{\Delta}(\frac{1}{2} \mathbf{1}_{[-1,1]})$ by Corollary 4.1. Melas [9] proved the sharper, optimal value $\|M_c\|_{1 \rightarrow 1, \infty} = \frac{11+\sqrt{61}}{12} = 1.5675208063\dots$ using discretization.

4.2 Euclidean balls

The centered Hardy–Littlewood maximal operator over Euclidean balls is

$$M_{B,n} f(x) = \sup_{r>0} \frac{1}{\omega_n r^n} \int_{B(x,r)} |f(y)| dy, \quad \omega_n = \frac{\pi^{n/2}}{\Gamma(n/2 + 1)}.$$

Here the averages are over solid Euclidean balls.

The classical covering argument gives an exponential bound 2^n [10, Exercise 42]. Stein–Strömberg improved the dimensional order to $O(n)$ [11].

Corollary 4.1 (Radial majorants). *Put*

$$\rho_n = \begin{cases} (n/2)^{2/(n-2)}, & n \neq 2, \\ e, & n = 2. \end{cases}$$

Suppose $\phi(x) = q(|x|^2)$, with q left-continuous at $a > 0$. If an admissible radial majorant for $\mathcal{A} = \Delta$ is harmonic for $0 < |x| < \sqrt{b}$, touches ϕ at $|x| = \sqrt{a}$, and equals ϕ for $|x| \geq \sqrt{b}$, where $b = \rho_n a$, then it attains $C_\Delta(\phi)$.

Proof. Rotational averaging preserves the mass and admissibility of any competitor. Write the average as $k(|x|^2)$. Subharmonicity means that k is convex in the coordinate

$$s(z) = \begin{cases} \frac{z^{1-n/2}}{1-n/2}, & n \neq 2, \\ \log z, & n = 2. \end{cases}$$

If Z has density $\frac{n}{2b^{n/2}} z^{n/2-1}$ on $(0, b)$, then $\mathbb{E}s(Z) = s(a)$. Jensen's inequality gives

$$\int_0^b k(z) z^{n/2-1} dz \geq \frac{2b^{n/2}}{n} k(a) \geq \frac{2b^{n/2}}{n} q(a).$$

Outside b use $k \geq q$. The stated majorant attains both inequalities, so no admissible kernel has smaller mass. $\square$

The following radial majorants give explicit constants:

$$\|M_{B,2}\|_{1 \rightarrow 1, \infty} \leq e, \quad \|M_{B,n}\|_{1 \rightarrow 1, \infty} \leq \left(\frac{n}{2}\right)^{n/(n-2)} \quad (n \geq 3).$$

For the planar unit ball use

$$K_2(x) = \frac{2}{\pi} \log \frac{\sqrt{e}}{|x|} \mathbf{1}_{\{|x| < \sqrt{e}\}}.$$

It majorizes $\pi^{-1} \mathbf{1}_{B(0,1)}$, its distributional Laplacian is a positive boundary measure minus an atom at zero, and its mass is e . For $n \geq 3$ put

$$R_n = \left(\frac{n}{2}\right)^{1/(n-2)}, \quad K_n(x) = \frac{1}{\omega_n} \frac{(|x|^{2-n} - R_n^{2-n})_+}{1 - R_n^{2-n}}.$$

The same conditions hold and direct polar integration gives

$$\int K_n = R_n^n = \left(\frac{n}{2}\right)^{n/(n-2)}.$$

These kernels attain $C_\Delta(\omega_n^{-1} \mathbf{1}_{\overline{B(0,1)}})$ by Corollary 4.1; changing the ball boundary does not affect the averages. The bound is asymptotic to $n/2$, whereas Spector–Stockdale prove $\|M_{B,n}\|_{1 \rightarrow 1, \infty} = O(\sqrt{n} \log(1+n))$ by exploiting the heat residual [1, Theorem 1.1].

4.3 Axis-parallel squares

The centered Hardy–Littlewood maximal operator over axis-parallel squares in the plane is

$$M_{\square}f(x) = \sup_{r>0} \frac{1}{4r^2} \int_{x+[-r,r]^2} |f(y)| dy.$$

The invertible map $(u, v) \mapsto (u + v, u - v)$ takes the unit diamond

$$D_{\diamond} = \{(u, v) : |u| + |v| \leq 1\}$$

to the unit square. Change of variables preserves a weak-type constant, and the diamond has area two. Use the homogeneous symmetric Markov generator

$$\mathcal{A}_{\alpha} = -|D_u|^{\alpha} - |D_v|^{\alpha}, \quad \widehat{\mathcal{A}_{\alpha}g}(\xi) = -(|\xi_1|^{\alpha} + |\xi_2|^{\alpha})\widehat{g}(\xi).$$

Corollary 3.1 gives

$$\|M_{\square}\|_{1 \rightarrow 1, \infty} \leq C_{\alpha} := \inf \left\{ \frac{1}{2} \int_{\mathbb{R}^2} K : K \in L^1(\mathbb{R}^2) \text{ is even,} \right. \\ \left. K \geq \mathbf{1}_{D_{\diamond}}, \quad \mathcal{A}_{\alpha}K \geq 0 \text{ on } \mathbb{R}^2 \setminus \{0\} \right\}.$$

Take $\alpha = 6/5$. A fractional radial profile on the diamond, corrected by a finite combination of cubic splines, gives an admissible kernel with

$$\boxed{\|M_{\square}\|_{1 \rightarrow 1, \infty} \leq C_{6/5} < 3.615749.}$$

Exact rational certificates verify domination and generator positivity [8]. This is a feasible comparison bound; attainment of the infimum $C_{6/5}$ is not established. For comparison, the covering bound is 4. A preceding certified bound 3.879 and a lower bound 1.68550999... are recorded in [optimizationproblems PR 192](#). The [GitHub development](#) and [Palomar v3](#) contain formalizations of that earlier square bound and the ball estimates. The refined bound 3.615749 is supported by the separate exact certificate in the accompanying supplement.

4.4 Poisson maximal

Define

$$P_*f(x) = \sup_{t>0} (p_t * |f|)(x), \quad p_t(x) = \frac{\Gamma((n+1)/2)}{\pi^{(n+1)/2}} \frac{t}{(t^2 + |x|^2)^{(n+1)/2}}.$$

Dimension-free weak $(1, 1)$ boundedness is a classical application of the Hopf–Dunford–Schwartz ergodic theorem [12, pp. 48–49]; see also [13, Theorem 5.5]. An explicit evaluation of this argument gives, in every dimension,

$$\|P_*\|_{1 \rightarrow 1, \infty} \leq B_P := \operatorname{erf} \sqrt{\frac{3}{2}} + \sqrt{\frac{3}{2\pi}} e^{-3/2} = 1.070915813140 \dots$$

Indeed, the subordination density

$$m_t(s) = \frac{t}{2\sqrt{\pi}} s^{-3/2} e^{-t^2/(4s)}$$

peaks at $s_0 = t^2/6$. Its nonincreasing majorant, constant before s_0 and equal to m_t thereafter, has mass B_P . Integration by parts dominates P_*f by B_P times the maximal heat ergodic average, whose weak constant is one.

The Laplacian majorant gives the following bound.

Corollary 4.2. *With ρ_n as in Corollary 4.1, let a be the unique nonzero solution in $(n/(3\rho_n), n/3)$ of*

$$\frac{1 - a/n}{(1 + a)^{(n+3)/2}} = \frac{1}{(1 + \rho_n a)^{(n+1)/2}}, \quad b = \rho_n a.$$

Then

$$\|P_*\|_{1 \rightarrow 1, \infty} \leq \frac{\Gamma((n+1)/2)}{\sqrt{\pi} \Gamma(n/2)} \left[\frac{2b^{n/2}}{n(1+a)^{(n+1)/2}} + \int_b^\infty \frac{z^{n/2-1}}{(1+z)^{(n+1)/2}} dz \right]. \quad (2)$$

For $n = 1$, this is

$$\|P_*\|_{1 \rightarrow 1, \infty} \leq 1 + \frac{2}{\pi} \left(\frac{\sqrt{5}}{3} - \arctan \frac{2}{\sqrt{5}} \right) = 1.009949307880 \dots$$

The right-hand side of (2) tends to $B_P = 1.070915813140 \dots$ as $n \rightarrow \infty$.

Proof. Write $\beta = n/2$, $q(z) = (1+z)^{-(n+1)/2}$, and $z = |x|^2$. For $z < b$, replace q by its radial harmonic tangent at a :

$$k(z) = \begin{cases} q(a) + \frac{aq'(a)}{1-\beta} ((z/a)^{1-\beta} - 1), & n \neq 2, \\ q(a) + aq'(a) \log(z/a), & n = 2. \end{cases}$$

For $z \geq b$, keep $k(z) = q(z)$. The equation for a makes the branches continuous. The Poisson profile has negative Laplacian for $z < n/3$ and positive Laplacian for $z > n/3$. Thus its graph is concave and then convex in the harmonic radial coordinate: the tangent majorizes the inner profile, and the derivative jumps upward at the outer joining point. Consequently

$$K(x) = \frac{\Gamma((n+1)/2)}{\pi^{(n+1)/2}} k(|x|^2) \geq p_1(x), \quad \Delta K = \nu - c\delta_0, \quad \nu \geq 0, \quad c > 0.$$

Corollary 3.1 bounds the maximal operator by $\int K$. Integrating the inner branch, using $\rho_n^{1-\beta} = 1/\beta$ when $n \neq 2$, gives

$$\int_0^b k(z) z^{\beta-1} dz = \frac{b^\beta q(a)}{\beta};$$

the logarithmic case gives the same identity. The majorant attains $C_\Delta(p_1)$ by Corollary 4.1. Polar integration now yields (2). In dimension one $a = 1/5$ and $b = 4/5$, which gives the stated elementary expression. Finally, $a/n \rightarrow 1/3$ and $b/n \rightarrow 1/3$. Stirling's formula gives the limit $\sqrt{3/(2\pi)}e^{-3/2}$ for the inner contribution. In the outer integral, the substitution $y = (n/2)/z$ gives the limiting contribution $\pi^{-1/2} \int_0^{3/2} y^{-1/2} e^{-y} dy = \operatorname{erf} \sqrt{3/2}$. $\square$

4.5 Heat maximal

Define

$$H_* f(x) = \sup_{t>0} (h_t * |f|)(x), \quad h_t(x) = (4\pi t)^{-n/2} e^{-|x|^2/(4t)}.$$

For each $n \geq 1$, use ρ_n from Corollary 4.1 and let $a = a(n)$ be the unique nonzero solution in $(n/(2\rho_n), n/2)$ of

$$e^{-(\rho_n-1)a} = 1 - \frac{2a}{n}.$$

Corollary 3.1 gives

$$\|H_*\|_{1 \rightarrow 1, \infty} \leq \frac{1}{\Gamma(n/2)} \left[\frac{2(\rho_n a)^{n/2} e^{-a}}{n} + \int_{\rho_n a}^{\infty} z^{n/2-1} e^{-z} dz \right].$$

To construct the majorant, put $z = |x|^2$ and replace $q(z) = e^{-z}$ for $0 < z < \rho_n a$ by its harmonic tangent at a :

$$k(z) = e^{-a} \begin{cases} 1 + \frac{2a}{n-2} ((z/a)^{1-n/2} - 1), & n \neq 2, \\ 1 - a \log(z/a), & n = 2. \end{cases}$$

For $z \geq \rho_n a$, keep $k(z) = e^{-z}$, and set $K(x) = \pi^{-n/2} k(|x|^2)$. The root equation makes the two branches meet. The profile is concave and then convex in the harmonic radial coordinate, because the sign of its Laplacian is the sign of $z - n/2$. The tangent lies above the inner profile and meets its convex outer branch with a nonnegative flux jump. Therefore $K \geq h_{1/4}$ and ΔK is a positive finite measure minus an atom at zero. Polar integration gives the displayed bound. The majorant attains $C_{\Delta}(h_{1/4})$ by Corollary 4.1.

In dimension one, $a(1)$ satisfies $e^{-3a(1)} = 1 - 2a(1)$, and the bound is

$$\|H_*\|_{1 \rightarrow 1, \infty} \leq 4 \sqrt{\frac{a(1)}{\pi}} e^{-a(1)} + \operatorname{erfc}(2\sqrt{a(1)}) = 1.037087059430 \dots,$$

where $\operatorname{erfc}(r) = \frac{2}{\sqrt{\pi}} \int_r^{\infty} e^{-s^2} ds$. In dimension two, $a(2)$ satisfies $e^{-(e-1)a(2)} = 1 - a(2)$, and

$$\|H_*\|_{1 \rightarrow 1, \infty} \leq [1 + (e-1)a(2)] e^{-a(2)} = 1.094052157640 \dots$$

For $n = 3$, the bound is 1.150845692298...

As $n \rightarrow \infty$,

$$a(n) = \frac{n}{2} - 1 + o(1), \quad \rho_n a(n) = \frac{n}{2} + \log \frac{n}{2} - 1 + o(1).$$

Stirling's formula shows that the resulting upper bound is asymptotic to $\sqrt{n}/(2\sqrt{\pi})$. This growth is worse than the $O(\log(1+n))$ heat bound proved by Spector–Stockdale [1, Theorem 1.2].

A Sharp planar bound for the second-order Riesz transform

We deduce the sharp planar matrix estimate from Guerra's quasiconvexity theorem [4, Theorem 1.1].

Corollary A.1. *For every real-valued $f \in L^1(\mathbb{R}^2)$ and $t > 0$,*

$$t |\{ |f| + |Bf| > t \}| \leq 2 \|f\|_1.$$

Consequently, $\|\mathcal{H}_2\|_{1 \rightarrow 1, \infty} = \|\mathcal{H}_{0,2}\|_{1 \rightarrow 1, \infty} = \sqrt{2}$.

Proof. Identify $\mathbb{R}^2$ with $\mathbb{C}$ and write a real matrix as $Az = a_+ z + a_- \bar{z}$. Then $\|A\|_{\text{op}} = |a_+| + |a_-|$ and $\det A = |a_+|^2 - |a_-|^2$. Guerra's integrand is

$$L(A) = \begin{cases} \det A, & |a_+| + |a_-| \leq 1, \\ 2|a_+| - 1, & |a_+| + |a_-| > 1. \end{cases}$$

Its definition gives the following strengthening of [4, (6.1)]:

$$L(A/t) \leq \frac{2|a_+|}{t} - \mathbf{1}_{\{|a_+|+|a_-|>t\}}.$$

Indeed, equality holds when $|a_+| + |a_-| > t$; otherwise the indicator vanishes and $\det(A/t) \leq |a_+|^2/t^2 \leq |a_+|/t$.

First take real $f \in C_c^\infty(\mathbb{R}^2)$ and set

$$u(x) = \frac{1}{\pi} \int_{\mathbb{R}^2} \log|x-y| f(y) dy, \quad \Delta u = 2f.$$

For $A = D^2u$, the complex coefficients satisfy

$$a_+ = \frac{u_{11} + u_{22}}{2} = f, \quad a_- = \frac{u_{11} - u_{22}}{2} + iu_{12}, \quad |a_-| = |Bf|.$$

We claim that Guerra's theorem implies

$$\int_{\mathbb{R}^2} L(D^2u/t) dx \geq 0.$$

To justify its use for this noncompact potential, choose a smooth cutoff η_R equal to one on $B(0, R)$ and zero outside $B(0, 2R)$, and apply the theorem at the zero matrix to $\eta_R u/t$. At infinity,

$$u = O(\log|x|), \quad \nabla u = O(|x|^{-1}), \quad D^2u = O(|x|^{-2}).$$

On the cutoff annulus, therefore, $D^2(\eta_R u) = O(R^{-2} \log R)$. For fixed t and sufficiently large R , this lies in the determinant branch of L , whose integral over that annulus is $O(R^{-2} \log^2 R) = o(1)$. Also $L(D^2u/t) = O(|x|^{-4})$ at infinity, so it is integrable. Letting $R \rightarrow \infty$ proves the claim.

Integrating the pointwise inequality now gives

$$0 \leq \int L(D^2u/t) \leq \frac{2}{t} \|f\|_1 - |\{|f| + |Bf| > t\}|.$$

Approximation in L^1 , using the weak-type continuity of B , extends this estimate to every real $f \in L^1$. By the identities in Section 2.3,

$$|\mathcal{H}_2 f|_F = \frac{\sqrt{|f|^2 + |Bf|^2}}{\sqrt{2}} \leq \frac{|f| + |Bf|}{\sqrt{2}}.$$

Taking $t = \sqrt{2}\lambda$ yields

$$\lambda |\{| \mathcal{H}_2 f|_F > \lambda \}| \leq \sqrt{2} \|f\|_1.$$

Conversely, $|\mathcal{H}_2 f|_F \geq |Bf|/\sqrt{2}$, so the sharp constant 2 for B on real inputs [4, Corollary 1.2] forces $\|\mathcal{H}_2\|_{1 \rightarrow 1, \infty} \geq \sqrt{2}$. Finally, $|\mathcal{H}_{0,2} f|_F = |Bf|/\sqrt{2}$ gives the same sharp constant for the traceless transform. $\square$

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